PHASE-2 · PROJECT IMPLEMENTATION · DL MINI PROJECT

IndicDoc Verify

Deep-learning layout analysis that checks Indian question papers and their translations **before they are printed**: offline, 12 languages

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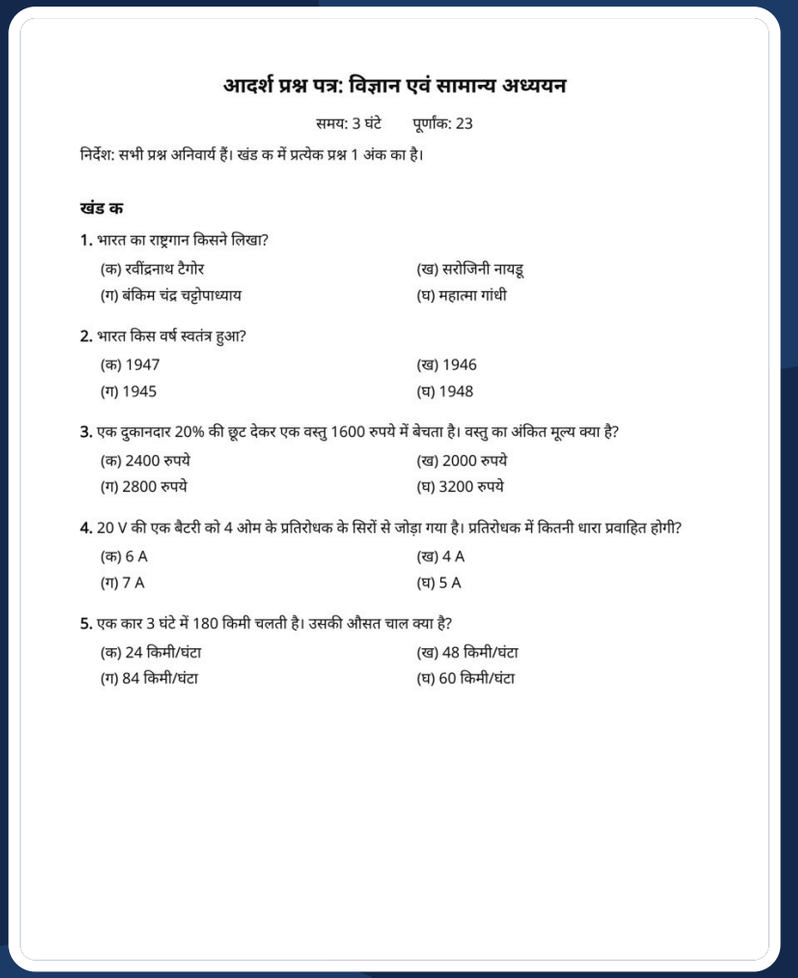
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Course: Deep Learning Mini Project (Expt. 11, SDG-based)

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आदर्श प्रश्न पत्र: विज्ञान एवं सामान्य अध्ययन

समय: 3 घंटे पूर्णांक: 23

निर्देश: सभी प्रश्न अनिवार्य हैं। खंड क में प्रत्येक प्रश्न 1 अंक का है।

खंड क

1. भारत का राष्ट्रगान किसने लिखा?
   (क) रवींद्रनाथ टैगोर (ख) सरोजिनी नायडू
   (ग) बंकिम चंद्र चट्टोपाध्याय (घ) महात्मा गांधी

2. भारत किस वर्ष स्वतंत्र हुआ?
   (क) 1947 (ख) 1946
   (ग) 1945 (घ) 1948

3. एक दुकानदार 20% की छूट देकर एक वस्तु 1600 रुपये में बेचता है। वस्तु का अंकित मूल्य क्या है?
   (क) 2400 रुपये (ख) 2000 रुपये
   (ग) 2800 रुपये (घ) 3200 रुपये

4. 20 V की एक बैटरी को 4 ओम के प्रतिरोधक के सिरों से जोड़ा गया है। प्रतिरोधक में कितनी धारा प्रवाहित होगी?
   (क) 6 A (ख) 4 A
   (ग) 7 A (घ) 5 A

5. एक कार 3 घंटे में 180 किमी चलती है। उसकी औसत चाल क्या है?
   (क) 24 किमी/घंटा (ख) 48 किमी/घंटा
   (ग) 84 किमी/घंटा (घ) 60 किमी/घंटा

OVERVIEW

Content

01 Literature Survey Summary Table

02 Research Gaps

03 Problem Statement and Objectives

04 Proposed System, Methodology and Novelty

05 System Block Diagram

06 Dataset Description and Pre-processing

07 Deep Learning Model Architecture

08 Tools, Technologies and Environment

09 Implementation Details

10 Experimental Setup and Training

11 Results and Performance Evaluation

12 Comparison with Baseline / Existing Methods

13 Sample Outputs and Demonstration

14 Challenges and Limitations

15 Conclusion and Future Scope

16 References

Literature Survey Summary Table

Ref	Author	Problem	Dataset	Method	Limitation
[1]	**Nath et al., 2025**	Layout parsing of Indian-language pages	IndicDLP: 119,806 pages, 12 languages, 42 labels incl. questions and options	Benchmark of YOLOv10x, DocLayout-YOLO, RoDLA, DINO, DiT, Florence-2	Detects regions only; nobody uses the question labels to check papers
[2]	**Kasuba et al., 2025**	Page-level handwritten Indic text recognition	CHIPS, ~25K pages, 10 languages	DBNet word detection + 6 recognisers (CRNN, PARSeq, ...)	User must name the language; text only
[3]	**Mondal et al., 2026**	Multilingual table detection	UniTabBank: 55K pages, 82K tables, 28 languages	Semi-automatic labelling; YOLOv11 detector	Borderless tables and degraded scans fail
[4]	**Pal et al., 2024**	Table question answering, low-resource Indic	BanglaTabQA 2M, HindiTabQA 643K	SQL-generated QA; fine-tuned mBART / M2M / Llama	Wikipedia tables only
[5]	**Xu et al., 2021**	Multilingual document understanding	30M docs; XFUND, 7 languages	LayoutXLM: text + layout + image transformer	No Indian language
[6]	**Kasuba et al., 2025**	Grounding answers on the page	MGVG: 70 circulars, 509 QA pairs	OCR + Llama-3.1-8B + region matching	OCR-dependent; single page
[7]	**Bhattacharyya et al., 2025**	Hindi OCR with vision-language models	1.1M printed Hindi words + pages	LoRA-tuned GOT-OCR 2.0	Hindi only; mixed-language lines are hard
[8]	**Dumada & Shah, 2026**	Devanagari table extraction	500+ Devanagari documents	Structure + Tesseract content, weighted fusion	No ablation; self-reported
[9]	**Babayeva et al., 2025**	Translating a test into other languages with an LLM	Force Concept Inventory, many languages	GPT-4o translation + back-translation + human review	Every item still checked by hand; no page structure
[10]	**Rei et al., 2022**	Translation quality estimation without a reference	WMT 2022 QE shared task	CometKiwi: multilingual encoder predicts quality per sentence	Plain sentences: no question numbers, options or marks

WHAT IS MISSING

Research Gaps

⚠ Exam papers are translated into up to 13 languages and vetted only by hand; errors reach candidates	→	✓ **An automatic pre-print check of every language version against the source** G1
⚠ Translation-quality tools [9, 10] read plain sentences: blind to question numbers, options, marks, figures	→	✓ **Compare versions question by question using the page layout itself** G2
⚠ Indic layout models [1] detect questions and options, but this is not used to check papers	→	✓ **Use the detector's question / option regions (AP@50 0.81–0.90) as the backbone of the checker** G3
⚠ Indic OCR misreads Latin digits in mixed text [7]; a checker of numbers needs every digit right	→	✓ **A "number pass": two OCR models must agree on every number** G4

We also looked at court-filing defect checks: the Supreme Court already uses an AI tool built with IIT Madras, and a commercial tool is reported in 16 High Courts, so we chose the problem **nobody automates yet**: checking a question paper and its translations.

WHAT WE SOLVE

Problem Statement and Objectives

Problem: a question paper is set in one language and translated into Hindi and regional languages. Every version is checked by hand before printing. When a number, an option, a mark or a whole question differs, lakhs of candidates are affected: grace marks, re-exams, court cases.

MAIN OBJECTIVE

Build an offline deep-learning system that reads a printed paper in any of 12 Indian languages and flags every structural or numerical error, and every difference from its source version, before printing.

SPECIFIC OBJECTIVES

- Detect the paper's structure (questions, options, answers) with a layout detector; compare 3 detectors.
- Read text with reliable numbers in 12 languages (number pass).
- Checks: numbering, duplicates, options, marks, translation parity, repeats, format profiles.
- Measure errors found and false alarms on a benchmark; ship app + API in Docker.

EXPECTED OUTPUT

A verdict (print / check / do not print) and each finding with its question and page, outlined on the page, as HTML and JSON.

IT HAS ALREADY HAPPENED

NEET-UG 2018
49 questions wrong in the Tamil version; HC ordered 196 grace marks, SC set it aside

KPSC KAS 2024
59 errors in the Kannada paper; re-exam for ~2 lakh candidates, errors again

APPSC Group-I
51 of 120 questions alleged mistranslated into Telugu; SC notice

UGC-NET Jun 2026
Translation errors + questions repeated from old papers; re-test 9–10 Sep

MBBS (NTRUHS) 2026
The same MCQ printed twice; AP HC ordered a compensatory mark

Sources: The News Minute, Deccan Herald, India Legal, The Week, SCC Online, Medical Dialogues (refs. slide).

OUR APPROACH

Proposed System, Methodology and Novelty

1. **Paper in**
   PDF / scans + source version
2. **Find structure**
   DocLayout-YOLO: questions, options
3. **Read text**
   Indic OCR + English number pass
4. **Build questions**
   number, stem, options, marks, figures
5. **Check**
   structure, parity, repeats, format
6. **Report**
   verdict + findings on the page

How deep learning is used

A convolutional object detector finds every question, sub-question, option and answer on the page (AP@50 0.81–0.90 on these labels). Its regions and the word positions inside them turn a picture of a page into structured questions, in any script.

What is new

Translation parity without translation: numbers, option counts, marks, sub-questions and figures must be the same in every language, so versions are compared question by question with no MT model.

Number pass: two OCR models must agree on every digit.

Why it sells

USP: no tool we found checks an Indian question paper against its translations question by question; IndicDoc Verify does, offline, in 12 languages, before printing. Confidential papers never leave the exam body. Same engine checks any document format via JSON profiles (e.g. bilingual circulars).

ARCHITECTURE

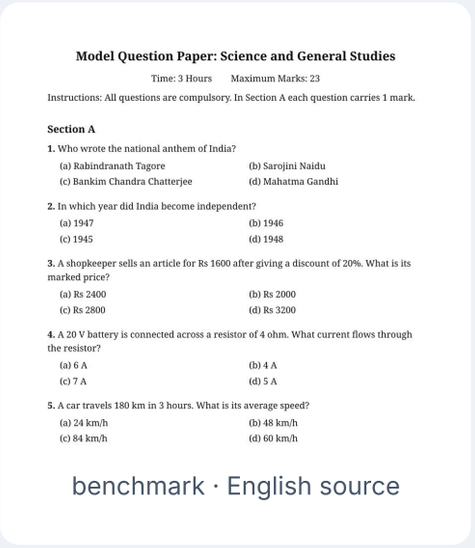
System Block Diagram

USER	**Streamlit: Verify Paper tab** upload paper, source, bank, profile	**REST API (FastAPI)** POST /verify · GET /verify/profiles	**Docker, offline** papers never leave the exam body
INPUT	**Render pages** PDF at 200 dpi, images	**Source version** optional, any of 12 languages	**Question bank · profile** earlier papers · JSON format rules
DEEP LEARNING	**Layout detector** DocLayout-YOLO (default) · YOLOv8s · Faster R-CNN	**Post-process** high-recall threshold, duplicate removal	**Question regions** questions, sub-questions, options, answers
TEXT	**Indic OCR** Tesseract, the page's own script	**Number pass** English model, digits must agree	**Questions** words → regions → options by layout gaps
CHECKS + OUTPUT	**Structure · repeats** numbering, duplicates, options, marks, bank	**Translation parity** align versions; numbers, options, marks	**Report** verdict, findings, pages outlined; HTML / JSON

DATA

Dataset Description and Pre-processing

Layout data	**IndicDLP** (Nath et al., ICDAR 2025; Hugging Face `ai4bharat/indicdlp`, AIKosh): 119,806 pages, 12 languages, 12 domains, 42 labels
Our subset	**3,317 pages**, 49,995 boxes, language-balanced, official splits: train 2,293 · val 342 · test 682. 195 question-paper pages; 1,227 question and 1,209 option boxes
Verify benchmark	**132 paper pairs** (English → Hindi, English → Tamil), 18 questions each (MCQ + short answer), rendered at 200 dpi; **108** with one injected error of 9 real kinds, 24 clean; half degraded like scans
Kept apart	A separate **development set** (48 pairs, other seed) was used to build the parser; the test set above was run once
Pre-processing	Parquet → PNG + COCO JSON; boxes clipped; COCO → YOLO labels; letterbox 640 / 1024 px. Verify: PDF → 200 dpi; blur 0.7 px + noise σ 8 + JPEG 55 for "scanned"
Augmentation	YOLOv8: mosaic, HSV, scale. Faster R-CNN: colour jitter, blur. No mirror flips
Challenges	Class imbalance ("paragraph" = 32% of boxes); no OCR ground truth in IndicDLP; no public set of erroneous exam papers, hence the benchmark
Deliverable	`dataset/` CSVs + samples; `scripts/verify_bench/` rebuilds the benchmark



Model Question Paper: Science and General Studies

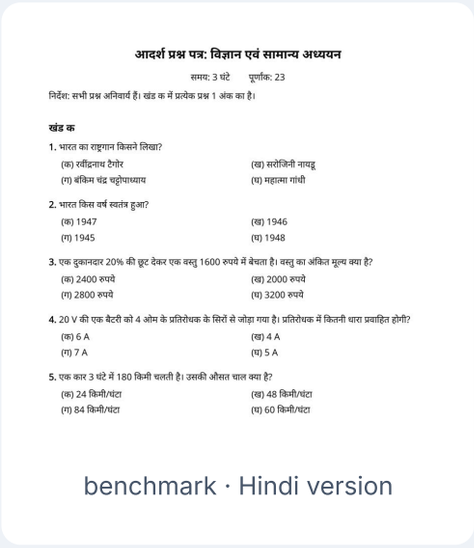
Time: 3 Hours Maximum Marks: 23

Instructions: All questions are compulsory. In Section A each question carries 1 mark.

Section A

1. Who wrote the national anthem of India?
   (a) Rabindranath Tagore (b) Sarojini Naidu
   (c) Bankim Chandra Chatterjee (d) Mahatma Gandhi
2. In which year did India become independent?
   (a) 1947 (b) 1946
   (c) 1945 (d) 1948
3. A shopkeeper sells an article for Rs 1600 after giving a discount of 20%. What is its marked price?
   (a) Rs 2400 (b) Rs 2000
   (c) Rs 2800 (d) Rs 3200
4. A 20 V battery is connected across a resistor of 4 ohm. What current flows through the resistor?
   (a) 6 A (b) 4 A
   (c) 7 A (d) 5 A
5. A car travels 180 km in 3 hours. What is its average speed?
   (a) 24 km/h (b) 48 km/h
   (c) 84 km/h (d) 60 km/h

benchmark · English source



आदर्श प्रश्न पत्र: विज्ञान एवं सामान्य अध्ययन

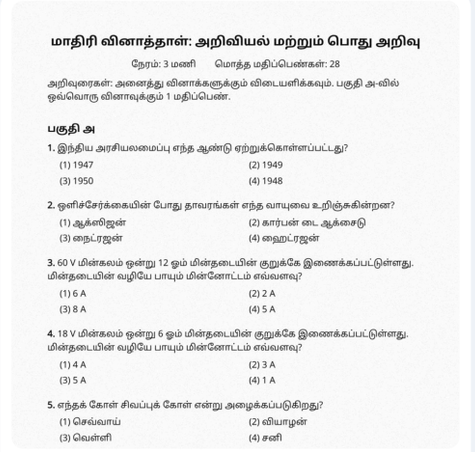
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निर्देश: सभी प्रश्न अनिवार्य हैं। खंड क में प्रत्येक प्रश्न 1 अंक का है।

खंड क

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   (ग) 84 किमी/घंटा (घ) 60 किमी/घंटा

benchmark · Hindi version



மாதிரி வினாத்தாள்: அறிவியல் மற்றும் பொது அறிவு

நேரம்: 3 மணி மொத்த மதிப்பெண்கள்: 28

அறிவுரைகள்: அனைத்து வினாக்களுக்கும் விடையளிக்கவும். பகுதி அ-வில் ஒவ்வொரு வினாவுக்கும் 1 மதிப்பெண்.

பகுதி அ

1. இந்திய அரசியலமைப்பு எந்த ஆண்டு ஏற்றுக்கொள்ளப்பட்டது?
   (1) 1947 (2) 1949
   (3) 1950 (4) 1948
2. ஒளிச்சேர்க்கையின் போது தாவரங்கள் எந்த வாயுவை உறிஞ்சுகின்றன?
   (1) ஆக்ஸிஜன் (2) கார்பன் டை ஆக்சைடு
   (3) நைட்ரஜன் (4) ஹைட்ரஜன்
3. 60 V மின்கலம் ஒன்று 12 Ω மின்தடையின் குறுக்கே இணைக்கப்பட்டுள்ளது. மின்தடையின் வழியே பாயும் மின்னோட்டம் எவ்வளவு?
   (1) 6 A (2) 2 A
   (3) 8 A (4) 5 A
4. 18 V மின்கலம் ஒன்று 6 Ω மின்தடையின் குறுக்கே இணைக்கப்பட்டுள்ளது. மின்தடையின் வழியே பாயும் மின்னோட்டம் எவ்வளவு?
   (1) 4 A (2) 3 A
   (3) 5 A (4) 1 A
5. எந்தக் கோள் சிவப்புக் கோள் என்று அழைக்கப்படுகிறது?
   (1) செவ்வாய் (2) வியாழன்
   (3) வெள்ளி (4) சனி

benchmark · Tamil, scan-degraded



8 जनसत्ता 11 अक्टूबर, 2023 देश

केसीआर ने गरीबों के लिए काम नहीं किया

'आप' विधायक अमानतुल्लाह के परिसरों पर ईडी के छापे

विशेषाधिकार समिति की बैठक में नहीं पहुंचे सांसद रमेश बिधूड़ी

आइएमएफ ने भारत की वृद्धि दर का अनुमान बढ़ाकर 6.3 फीसद किया

कैदियों की समय से पहले रिहाई मामले में सुप्रीम कोर्ट ने गुजरात सरकार से पूछा माफी की मांग पर निर्णय लेने में कितना समय लगता है?

राजस्थान में महिला निशानेबाजों से बलात्कार व छेड़छाड़, प्रशिक्षक के खिलाफ प्राथमिकी

संजय सिंह की ईडी हिरासत 13 अक्तूबर तक बढ़ी

IndicDLP · hindi newspaper

MODELS

Deep Learning Model Architecture

DocLayout-YOLO · 20.0 M params · used by IndicDoc Verify

Input	Backbone	Neck	Head	Output
1024×1024×3 page	CSP conv blocks + GL-CRM (global-to-local receptive fields)	PAN feature pyramid, 3 scales	YOLOv10-style, NMS-free	N × (box, 42-class score)

Fine-tuned on full IndicDLP by its authors (Zenodo 15881917). Question labels: first-level 0.85, second-level 0.90, options 0.84, answer 0.81 AP@50.

YOLOv8s · 11.2 M params · fine-tuned by us

Input	Backbone	Neck	Head	Output
640×640×3	CSPDarknet with C2f blocks + SPPF	PAN-FPN	decoupled, anchor-free	boxes + 42 classes

Transfer learning: COCO-pretrained weights, 80-class head replaced by a 42-class head. Loss: CIoU + DFL (boxes) + BCE (classes).

Faster R-CNN · 41.5 M params · baseline

Input	Backbone	Neck	RPN + RoIAlign	Box head
longest side 640–1024	ResNet-50, frozen batch-norm	FPN, 5 levels	region proposals	43 outputs (42 + background)

Two-stage detector, COCO-pretrained; predictor replaced (43 classes). Loss: RPN + classification CE + smooth-L1 box.

STACK

Tools, Technologies and Environment

Component	Details
Programming language	Python 3.11 (ML, checks, app, API); JavaScript (Node) for the benchmark renderer
DL framework	PyTorch 2 with torchvision; Ultralytics 8 (YOLOv8); doclayout-yolo 0.0.4
Development environment	VS Code / terminal, Git + GitHub, pytest (unit tests for every check)
Libraries	pycocotools, OpenCV, Pillow, NumPy, pandas, Matplotlib, PyMuPDF; Tesseract 5 (tessdata_best, 12 scripts); tesseract.js for the benchmark
Hardware	Apple M1 Pro GPU via PyTorch MPS (training, evaluation); CPU container for the benchmark; CPU in Docker
Dataset tools	Hugging Face Hub + fsspec (partial Parquet reads); COCO / YOLO converters; Playwright + Noto fonts to render benchmark papers
Interface / deployment	Streamlit app (Verify Paper tab) + FastAPI (/verify); Docker + docker-compose, offline

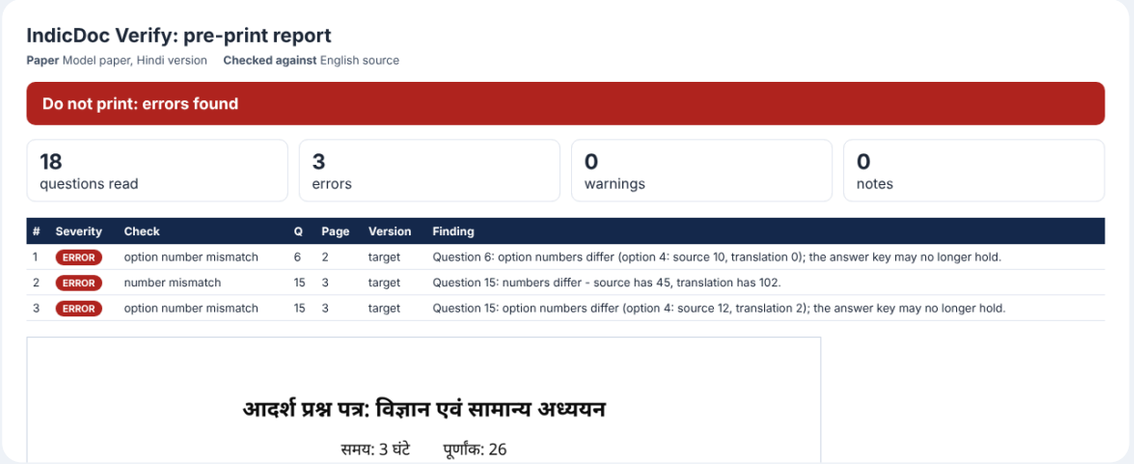
WHAT WE BUILT

Implementation Details

- **Layout model:** COCO-pretrained YOLOv8s and Faster R-CNN fine-tuned (SGD 0.005 / auto, CIoU+DFL+BCE), DocLayout-YOLO used for verification at its high-recall threshold (0.15).
- **OCR:** the page is read once with its own script's model; an English **number pass** fixes misread Latin digits (Hindi reads "1947" as "947") and the two must agree on every number.
- **Questions:** question regions start a question; options are separated by **layout gaps** between words, so garbled markers do not matter; marks, figures, tables attached.
- **Checks:** numbering, duplicates, option count, marks total; **parity** by global alignment (Needleman–Wunsch) on language-independent features; repeats by character 3-gram similarity; format profiles in JSON.
- **Output:** verdict, findings with question and page, problems outlined on the page; HTML / JSON; same code in app, API and benchmark.

```
# every page: layout (DL) -> OCR with number pass
out = predict_single(model, page, "doclayout", device, conf=0.15)
doc = extract_document(page, out, "hin", number_pass=True)

# ml/verify — questions, checks, parity, repeats
report = verify(hindi_pages, source_pages=english_pages,
                bank=earlier_questions, profile="question_paper")
report["summary"]   # {questions, errors, warnings}
html = to_html(report, {"paper": page_images})
```



IndicDoc Verify: pre-print report

Paper Model paper, Hindi version **Checked against** English source

Do not print: errors found

18	3	0	0
questions read	errors	warnings	notes

#	Severity	Check	Q	Page	Version	Finding
1	ERROR	option number mismatch	6	2	target	Question 6: option numbers differ (option 4: source 10, translation 0); the answer key may no longer hold.
2	ERROR	number mismatch	15	3	target	Question 15: numbers differ - source has 45, translation has 102.
3	ERROR	option number mismatch	15	3	target	Question 15: option numbers differ (option 4: source 12, translation 2); the answer key may no longer hold.

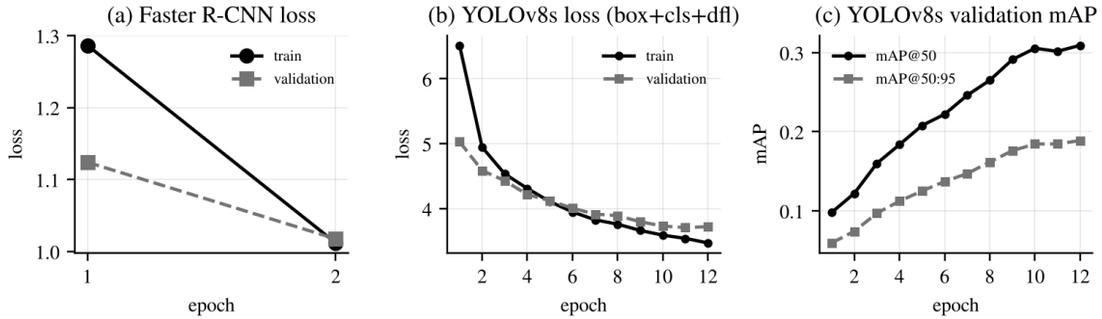
आदर्श प्रश्न पत्र: विज्ञान एवं सामान्य अध्ययन

समय: 3 घंटे पूर्णांक: 26

EXPERIMENTS

Experimental Setup and Training

Layout training	train 2,293 · val 342 · test 682 pages. Faster R-CNN: batch 4, SGD 0.005, **2 epochs, 76 min**. YOLOv8s: 640 px, batch 8, **12 epochs, 60 min**. Apple M1 Pro GPU
E1 Layout	3 detectors on 682 test pages: COCO mAP@50, mAP@50:95, P / R / F1, per label (incl. question labels)
E2 Robustness	blur, noise, JPEG, half resolution, faded print (Robustness Index)
E3 Verify benchmark	132 pairs, one injected error or none; layout boxes from the renderer, **real Tesseract OCR**; an error counts as found only if a matching finding points at the right question
E4 Ablations	without the number pass; without layout-based options; with a **noisy layout** (boxes ±8 px, 10% of question / option labels wrong)
Protocol	parser built on a separate development set; test set run once



Training curves of our two detectors (measured).

RESULTS (1/2) · LAYOUT

Results and Performance Evaluation

Model (682 test pages)	mAP@50	mAP@50:95	Precision*	Recall*	F1*	Recall, high-recall mode	s / page
DocLayout-YOLO (authors)	0.759	0.575	0.848	0.775	0.810	0.851	0.150
YOLOv8s (ours)	0.259	0.165	0.603	0.508	0.551	0.508	0.024
Faster R-CNN (ours)	0.148	0.079	0.510	0.435	0.469	0.435	0.202

* at the "balanced" threshold chosen on validation pages. Apple M1 Pro GPU.

AP@50 on question labels	DocLayout-YOLO	YOLOv8s	Faster R-CNN
first-level-question	0.854	0.415	0.240
second-level-question	0.898	0.045	0.018
third-level-question	0.872	0.026	0.000
options	0.839	0.553	0.243
answer	0.809	0.000	0.000

The full-data model finds the parts of a question paper reliably; this is what the checker is built on.

0.759

mAP@50, best model; question-paper domain 0.84

97%

of accuracy kept on damaged scans (mean Robustness Index)

Results and Performance Evaluation

98%

of injected errors found *at the right question* on born-digital papers (59 errors); 98% on scan-degraded pages

3.0

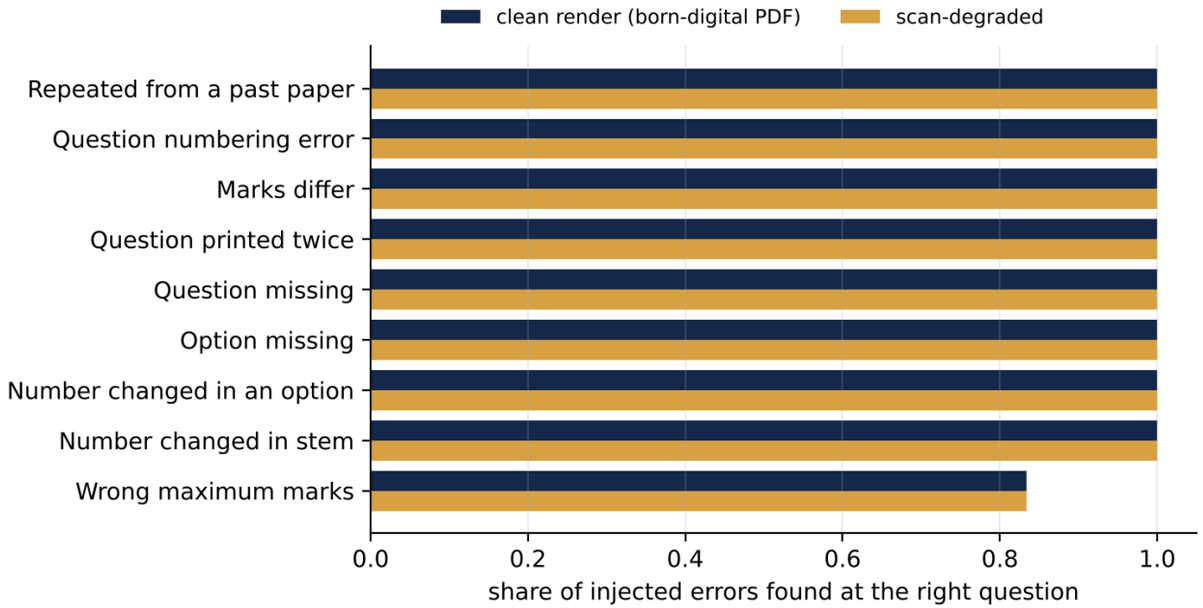
false alarms per clean paper pair, born-digital (5.5 when scan-degraded); each is shown on the page for the vetter to dismiss

98%

still found with a noisy layout (boxes shifted, 10% of question / option labels wrong)

132/132

papers with every question recovered from the pages (Hindi and Tamil)



Share of injected errors found at the right question, per error type (132 paper pairs, Hindi and Tamil).

COMPARISON

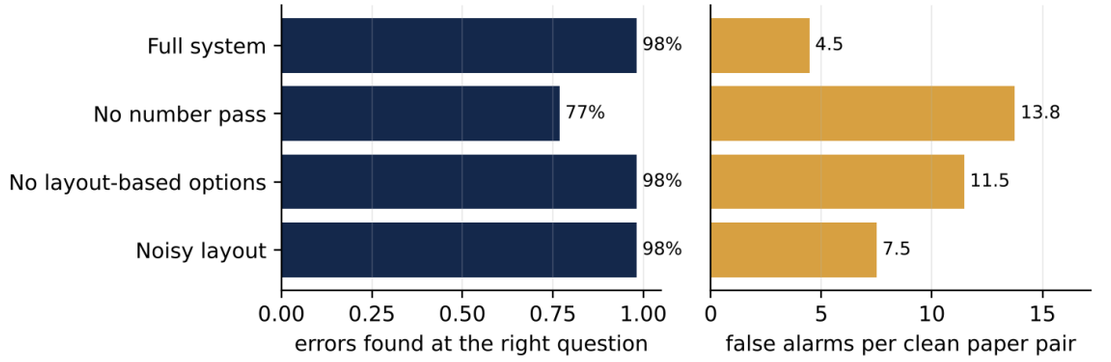
Comparison with Baseline / Existing Methods

Layout detector	Training data	mAP@50	mAP@50:95
Faster R-CNN (ours, baseline)	2,293 pages, 76 min	0.148	0.079
YOLOv8s (ours)	2,293 pages, 60 min	0.259	0.165
DocLayout-YOLO (authors)	~96K pages	**0.759**	**0.575**
Published: DocLayout-YOLO [1]	*full test set*	*≈0.735*	*0.545*

YOLOv8s beats the Faster R-CNN baseline at 9× the speed; the released full-data model is best and is used.

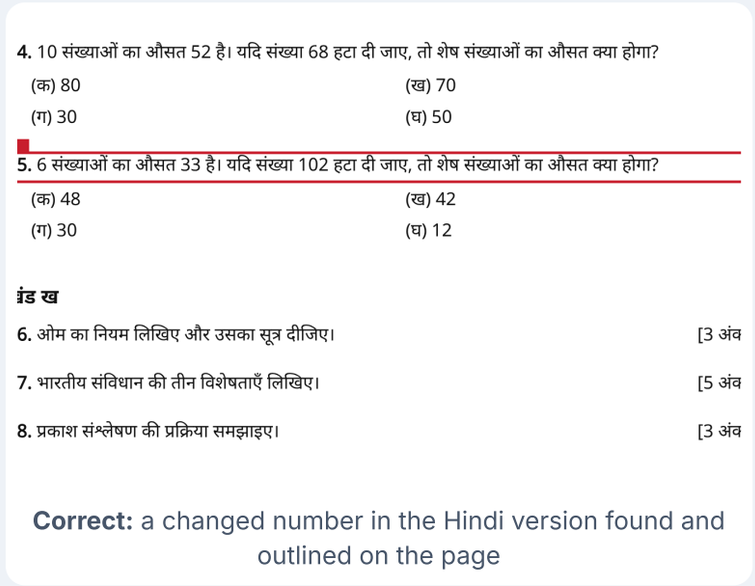
Verification ablation (test set)	found	false alarms / clean pair (all pages)
Full system	98%	4.5
Without number pass	77%	13.8
Without layout-based options	98%	11.5
Noisy layout	98%	7.5

Existing approach	Checks the paper?	Compares versions?	Indic, offline?
Manual vetting (status quo)	yes, by hand	by hand	yes
Answer-sheet AI (Eklavvya, GradeLab)	no, grades answers	no	partly
MT quality estimation (CometKiwi [10])	no structure	sentences only	cloud / GPU
LLM back-translation [9]	no structure	with human review	cloud
IndicDoc Verify	**numbering, options, marks, repeats**	**question by question**	**12 languages, offline**



DEMO

Sample Outputs and Demonstration



4. 10 संख्याओं का औसत 52 है। यदि संख्या 68 हटा दी जाए, तो शेष संख्याओं का औसत क्या होगा?

(क) 80 (ख) 70

(ग) 30 (घ) 50

5. 6 संख्याओं का औसत 33 है। यदि संख्या 102 हटा दी जाए, तो शेष संख्याओं का औसत क्या होगा?

(क) 48 (ख) 42

(ग) 30 (घ) 12

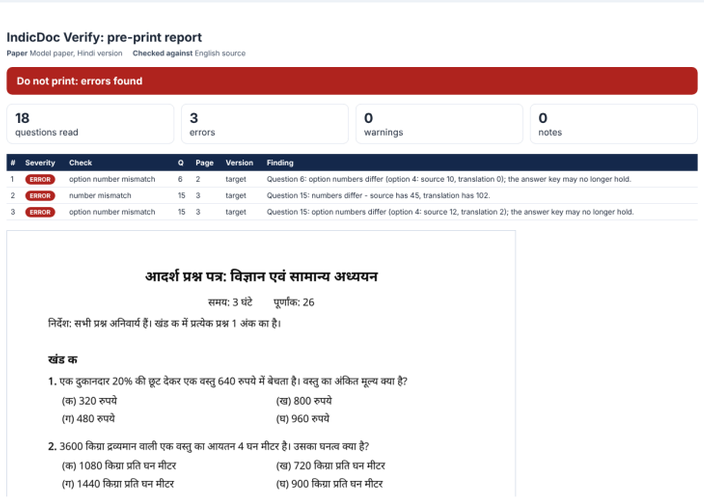
ंड ख

6. ओम का नियम लिखिए और उसका सूत्र दीजिए। [3 अंव

7. भारतीय संविधान की तीन विशेषताएँ लिखिए। [5 अंव

8. प्रकाश संश्लेषण की प्रक्रिया समझाइए। [3 अंव

Correct: a changed number in the Hindi version found and outlined on the page



IndicDoc Verify: pre-print report

Paper Model paper, Hindi version Checked against English source

Do not print: errors found

18	3	0	0
questions read	errors	warnings	notes

#	Severity	Check	Q	Page	Version	Finding
1	ERROR	option number mismatch	6	2	target	Question 6: option numbers differ (option 4: source 10, translation 0); the answer key may no longer hold.
2	ERROR	number mismatch	15	3	target	Question 15: numbers differ - source has 45, translation has 102.
3	ERROR	option number mismatch	15	3	target	Question 15: option numbers differ (option 4: source 12, translation 2); the answer key may no longer hold.

आदर्श प्रश्न पत्र: विज्ञान एवं सामान्य अध्ययन

समय: 3 घंटे पूर्णांक: 26

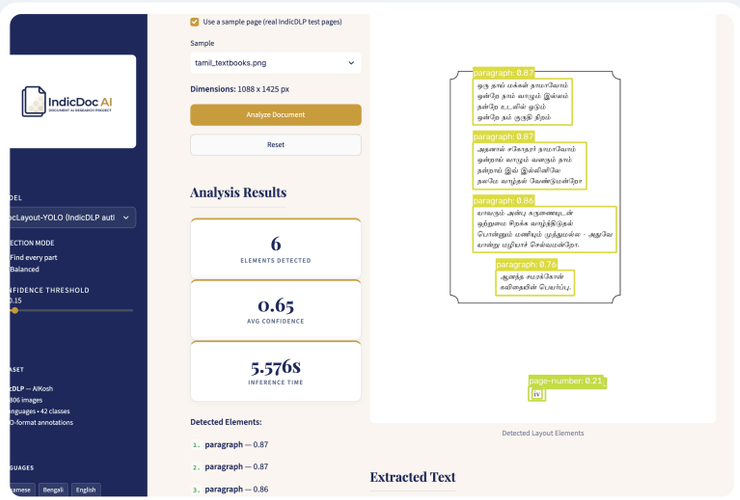
निर्देश: सभी प्रश्न अनिवार्य हैं। खंड क में प्रत्येक प्रश्न 1 अंक का है।

खंड क

1. एक दुकानदार 20% की छूट देकर एक वस्तु 640 रुपये में बेचता है। वस्तु का अंकित मूल्य क्या है?
(क) 320 रुपये (ख) 800 रुपये
(ग) 480 रुपये (घ) 960 रुपये

2. 3600 किग्रा द्रव्यमान वाली एक वस्तु का आयतन 4 घन मीटर है। उसका घनत्व क्या है?
(क) 1080 किग्रा प्रति घन मीटर (ख) 720 किग्रा प्रति घन मीटर
(ग) 1440 किग्रा प्रति घन मीटर (घ) 900 किग्रा प्रति घन मीटर

Verification report: verdict, findings with question and page (HTML / JSON)



IndicDoc AI

Use a sample page (real IndicDLP test pages)

Sample

tamil_textbooks.png

Dimensions: 1088 x 1425 px

Analyze Document

Reset

Analysis Results

6 ELEMENTS DETECTED

0.65 AVG CONFIDENCE

5.576s INFERENCE TIME

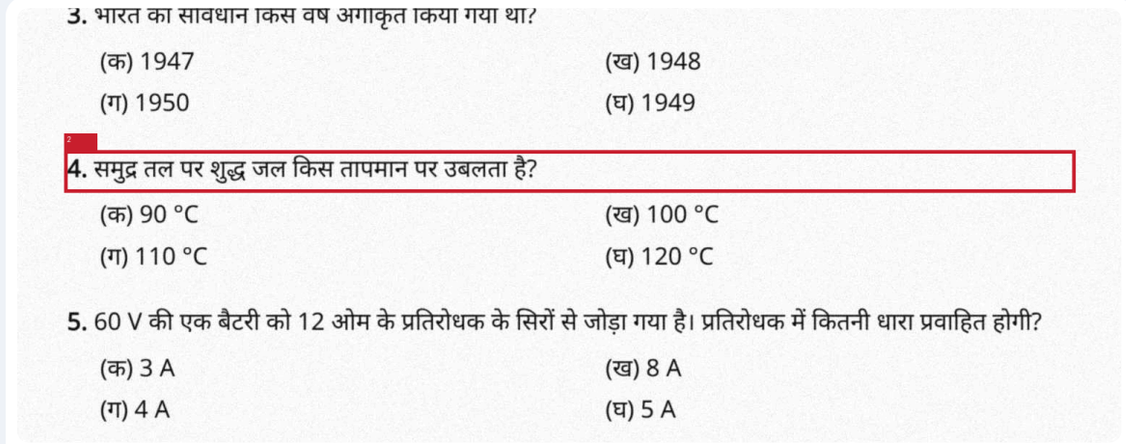
Detected Elements:

1. paragraph — 0.87
2. paragraph — 0.87
3. paragraph — 0.86

Extracted Text

Detected Layout Elements

Layout model in the running app: regions of a Tamil page



3. भारत का सावधान किस वष अगाकृत किया गया था?

(क) 1947 (ख) 1948

(ग) 1950 (घ) 1949

4. समुद्र तल पर शुद्ध जल किस तापमान पर उबलता है?

(क) 90 °C (ख) 100 °C

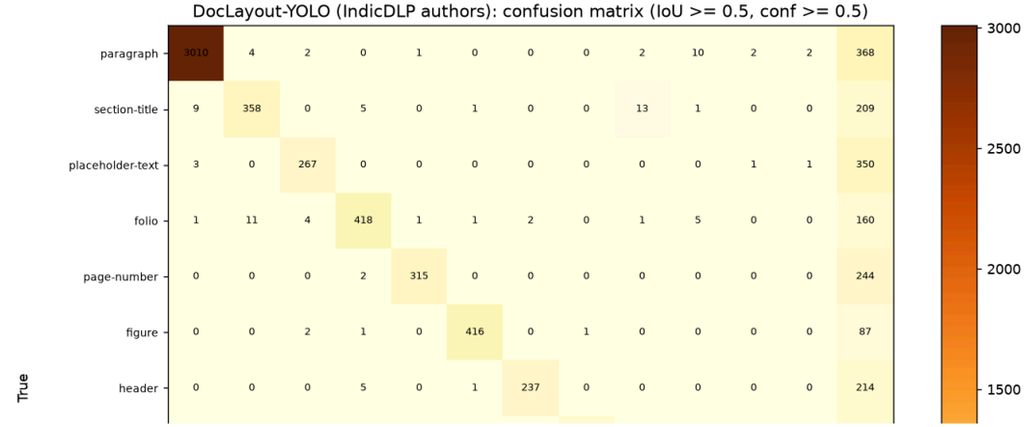
(ग) 110 °C (घ) 120 °C

5. 60 V की एक बैटरी को 12 ओम के प्रतिरोधक के सिरों से जोड़ा गया है। प्रतिरोधक में कितनी धारा प्रवाहित होगी?

(क) 3 A (ख) 8 A

(ग) 4 A (घ) 5 A

Incorrect: a false alarm on a degraded scan: OCR lost a digit of an option, so the versions look different



Layout model confusion matrix: most errors are missed regions, not wrong labels

LESSONS

Challenges and Limitations

OCR is the weakest link

Indic Tesseract models misread Latin digits ("1947" → "947", "1" → "।") and glue option markers to values. The number pass, layout-based option splitting and page-level digit-script detection were needed to get false alarms down.

No public data of real errors

Exam bodies do not publish erroneous pre-print papers, so we built a benchmark with realistic injected errors, kept a separate development set, and report clean and scan-degraded results separately.

Meaning is not checked yet

Parity catches numbers, options, marks and missing questions without translation. A pure change of meaning (a dropped word) needs a cross-lingual model; there is a hook for one, not yet evaluated.

Compute and data size

Our own detectors saw 2.3K pages for an hour (0.26 vs 0.76 mAP@50 for the released model); the product therefore uses the released model, and our work is the checker built on it.

WRAP-UP

Conclusion and Future Scope

CONCLUSION

- **Implemented:** IndicDoc Verify — layout detection (3 detectors compared), OCR for 12 scripts with a number pass, question parsing, structure / parity / repeat / format checks, report, web app, REST API, Docker.
- **Results:** 98% of injected errors found at the right question on born-digital papers (98% on degraded scans), 3.0 false alarms per clean born-digital pair; best layout model 0.759 mAP@50.
- **Objectives:** all four met. Our own detectors did not reach the released model, which we report openly.

SELLING IT

- **Buyers:** NTA, UPSC and state PSCs, school boards, university exam cells, test-series publishers; offline licence per exam body.

FUTURE SCOPE

- Pilot with a university exam cell or a state board on real pre-print papers; measure minutes saved per paper.
- Meaning check with a cross-lingual sentence model; number words and formulas.
- Better Indic OCR behind the same interface; answer-key consistency check.
- More document profiles: bilingual government circulars (Official Languages Act s.3(3)), forms, certificates.

SDG 4 | SDG 9 | SDG 10 | SDG 16

SOURCES

References

1. O. Nath, S. Kukkala, M. Khapra, R. K. Sarvadevabhatla, "IndicDLP: A Foundational Dataset for Multi-Lingual and Multi-Domain Document Layout Parsing," ICDAR 2025.
2. B. V. Kasuba et al., "PLATTER: A Page-Level Handwritten Text Recognition System for Indic Scripts," arXiv:2502.06172, 2025.
3. A. Mondal et al., "UniTabBank: A Large Scale Multi-Lingual, Multi-Layout, Multi-Type, Multi-Format Dataset for Table Detection," WACV 2026.
4. V. Pal, E. Kanoulas, A. Yates, M. de Rijke, "Table Question Answering for Low-resourced Indic Languages," arXiv:2410.03576, 2024.
5. Y. Xu et al., "LayoutXLM: Multimodal Pre-training for Multilingual Visually-rich Document Understanding," arXiv:2104.08836, 2021.
6. B. V. Kasuba, P. Chaudhuri, G. Ramakrishnan, "DRISHTIKON: Visual Grounding at Multiple Granularities in Documents," arXiv:2506.21316, 2025.
7. S. Bhattacharyya et al., "Adapting Vision-Language Models for Hindi OCR," IIIT Hyderabad, 2025.
8. A. R. Dumada, S. G. Shah, "Incorporate Structure and Content for Devanagari Table Extraction," IJSAT 17(1), 2026.
9. M. Babayeva, J. Dunlap, M. Snětinová, R. Widenhorn, "Translating the Force Concept Inventory in the Age of AI," arXiv:2508.13908, 2025.
10. R. Rei et al., "CometKiwi: IST-Unbabel 2022 Submission for the Quality Estimation Shared Task," WMT 2022.
11. Z. Zhao et al., "DocLayout-YOLO: Enhancing Document Layout Analysis through Diverse Synthetic Data and Global-to-Local Adaptive Perception," arXiv:2410.12628, 2024.
12. Y. Chen et al., "RoDLA: Benchmarking the Robustness of Document Layout Analysis Models," CVPR 2024.
13. S. Verma et al., "MILU: A Multi-task Indic Language Understanding Benchmark," NAACL 2025.
14. G. Jocher, A. Chaurasia, J. Qiu, "Ultralytics YOLOv8," 2023.
15. J. Ansel et al., "PyTorch 2: Faster Machine Learning Through Dynamic Python Bytecode Transformation and Graph Compilation," ASPLOS 2024.
16. S. Ren, K. He, R. Girshick, J. Sun, "Faster R-CNN," NeurIPS 2015 (the baseline architecture; the only pre-2021 source).

OTHER SOURCES

Dataset: huggingface.co/datasets/ai4bharat/indicdlp (AIKosh)

Weights: zenodo.org/records/15881917

Tesseract OCR 5 (tessdata_best), tesseract.js 7

Cases: The News Minute (NEET 2018), Deccan Herald / ETV Bharat (KPSC 2024-25), India Legal (APPSC), The Week / SCC Online (UGC-NET 2026), Medical Dialogues (MBBS 2026)

Code: github.com/va-ada/IndicDoc-AI

Thank You

Questions and discussion

Prashant Jha | Aditya Soni | Riddhi Patil | Mariam Badure

IndicDoc AI | Phase-2: Project Implementation